

Business Problem Statement :

A leading retail company aims to gain deeper insights into customer shopping behaviour to enhance sales, improve customer satisfaction, and increase long-term loyalty. Management has observed noticeable shifts in purchasing patterns across different demographic groups, product categories, and sales channels (online and offline). They want to understand which key factors—such as discounts, product reviews, seasonal trends, or payment preferences—have the strongest influence on customer decisions and repeat purchases.

You are assigned to analyze the company's consumer behaviour dataset to address the central business question:

“How can the company utilize consumer shopping data to identify trends, strengthen customer engagement, and optimize marketing and product strategies?”

Deliverables:

1. Data Preparation & Processing (Python)

Clean, preprocess, and transform the raw dataset into an analysis-ready format.

2. Data Analysis (SQL)

Organize the cleaned data into structured tables, simulate transactional scenarios, and run SQL queries to uncover insights related to customer segments, loyalty patterns, and purchase drivers.

3. Visualization & Insights (Power BI)

Develop an interactive Power BI dashboard that highlights essential trends and patterns, enabling business stakeholders to make informed, data-driven decisions.

4. Report & Presentation

Create a clear and concise project report summarizing key insights, findings, and business recommendations. Prepare a visual presentation that communicates these insights effectively to stakeholders.